

TUTORIAL – 8: Bugzilla & TestLink

Aim

To apply **defect management**, **test case management**, and **test execution report generation** using **Bugzilla** and **TestLink** workflows.

Description

Software testing tools help testers efficiently manage **test cases**, **test execution**, and **software defects**.

Bugzilla

Bugzilla is an open-source **Defect Tracking Tool** used to:

- Report software bugs (defects)
- Assign bugs to developers
- Track bug status
- Maintain bug history
- Generate defect reports

Common Bug Status

- New
- Assigned
- Open
- In Progress
- Resolved
- Verified
- Closed

TestLink

TestLink is an open-source **Test Management Tool** used to:

- Create Test Projects
- Create Test Plans
- Create Test Suites
- Design Test Cases
- Execute Test Cases
- Generate Test Reports
- Maintain Requirement Traceability

Bugzilla Workflow

Tester



Report Bug



New



Assigned



Developer Fixes Bug



Resolved



Retesting



Verified



Closed

TestLink Workflow

Create Project



Create Test Plan



Create Test Suite



Create Test Cases



Execute Test Cases



Generate Reports

Problem Statement

Manage test cases and software defects for the **Login Module** of the **Student Management System** using **TestLink** for test management and **Bugzilla** for defect tracking. Execute the test cases and generate Test Execution and Test Summary Reports.

How to Perform This Practical (Student Procedure)

Part A – Test Case Management Using TestLink

Step 1: Open TestLink

Login using:

- Username
- Password

Step 2: Create a Test Project

Project Name:

Student Management System

Step 3: Create Test Plan

Test Plan Name:

Login Module Testing

Step 4: Create Test Suite

Test Suite Name:

Authentication Testing

Step 5: Create Test Cases

Example:

TC001

Scenario:

Valid Login

Input:

admin/admin123

Expected Result:

Login Successful

Step 6: Execute Test Cases

Run all test cases.

Record:

- Pass
- Fail

Part B – Defect Management Using Bugzilla

Step 1: Login to Bugzilla

Enter

Username

Password

Step 2: Create New Bug

Fill

- Product
- Component
- Version
- Severity
- Priority
- Summary

Step 3: Enter Bug Details

Example

Bug Summary

"Blank username accepted"

Step 4: Save Bug

Bugzilla generates

Example

BUG-001

Step 5: Assign Bug

Assign to Developer.

Step 6: Retest

After developer fixes the bug,

Retest

Status
Verified
Closed

Implementation Details

The **Student Management System** Login Module is tested using **TestLink** for managing test cases and **Bugzilla** for recording and tracking defects. Test execution reports and summary reports are generated after executing all test cases.

Parameter Details

Parameter	Details
Project Name	Student Management System
Application/Software	Student Management System
Module Name	Login Module
SDLC Model	Waterfall Model
Programming Language	Python
IDE/Tool Used	Visual Studio Code, TestLink, Bugzilla
Input/Test Data	Username: admin Password: admin123

Source Code

```
username = input("Enter Username: ")
password = input("Enter Password: ")

if username == "admin" and password == "admin123":
    print("Login Successful")
else:
    print("Invalid Username or Password")
```

Test Case Detailed

Field	Details
Test Case ID	TC001
Requirement ID	R001
Module Name	Login Module
Feature Under Test	User Authentication
Test Scenario	Login using valid username and password
Test Objective	Verify successful login
Test Type (Unit/Integration/System/UAT)	System Testing
Test Technique (Black Box/White Box)	Black Box Testing
Priority	High
Severity	Critical
Preconditions	Login page is displayed
Postconditions	User enters Home Page
Test Environment	Windows 10
Browser/OS (if applicable)	Google Chrome
Test Data	admin / admin123
Expected Result	Login Successful
Actual Result	Login Successful

Status (Pass/Fail)	Pass
Executed By	Student Name
Execution Date	DD/MM/YYYY

Test Case Execution

Step No.	Test Steps	Input	Expected Result	Actual Result	Status
1	Open Login Page	-	Login page displayed	Login page displayed	Pass
2	Enter Username	admin	Username accepted	Username accepted	Pass
3	Enter Password	admin123	Password accepted	Password accepted	Pass
4	Click Login	Click	Login Successful	Login Successful	Pass

Testing Activities Performed

Sr. No.	SDLC Phase	STLC Phase	Testing Activity	Deliverable/Artifact
1	Requirement Analysis	Requirement Analysis	Review SRS	SRS Document
2	Design	Test Planning	Create Test Plan in TestLink	Test Plan
3	Development	Test Case Development	Create Test Cases	Test Case Document
4	Testing	Test Execution	Execute Test Cases in TestLink	Test Execution Report

5	Testing	Defect Reporting	Report Defects in Bugzilla	Bug Report
---	---------	------------------	----------------------------	------------

Test Execution Report

Test Case ID	Module Name	Test Scenario	Expected Result	Actual Result	Status	Remarks
TC001	Login	Valid Login	Login Successful	Login Successful	Pass	No Defect
TC002	Login	Invalid Password	Error Message	Error Message	Pass	Validation Correct
TC003	Login	Blank Username	Username Required	Username Required	Pass	Validation Correct
TC004	Login	Blank Password	Password Required	Password Required	Pass	Validation Correct

Test Summary Report

Metric	Value
Total Test Cases Designed	4
Total Test Cases Executed	4
Total Passed Test Cases	4
Total Failed Test Cases	0
Total Blocked Test Cases	0
Total Not Executed Test Cases	0
Pass Percentage (%)	100%

Fail Percentage (%)	0%
Overall Test Status	Pass
Tester Name	Student Name
Execution Date	DD/MM/YYYY

Observation

Sr. No.	Observation	Remarks
1	Test cases created successfully in TestLink	Completed
2	Test execution performed successfully	Pass
3	Defects can be reported in Bugzilla	Completed
4	Test reports generated successfully	Completed
5	No critical defects found	Pass

Output

TESTLINK

Project Created

Student Management System

Test Plan Created

Login Module Testing

Test Cases Executed

Total : 4

Passed : 4

Failed : 0

BUGZILLA

Bug Report Created

Bug ID : BUG-001

Status : New

Assigned : Developer

After Fix

Retest : PASS

Status : Closed

Testing Completed Successfully

Solve the Below Mention Questionnaires

1. What is Bugzilla primarily used for?

- A) Source Code Development
- B) Defect Tracking and Management**
- C) Database Design
- D) UI Design

Answer: B) Defect Tracking and Management

2. What is TestLink primarily used for?

- A) Network Monitoring
- B) Test Case Management**
- C) Database Testing
- D) Performance Monitoring

Answer: B) Test Case Management

3. Which status appears first in the Defect Life Cycle?

- A) Assigned
- B) Open
- C) New**
- D) Closed

Answer: C) New

4. Which document provides overall testing statistics and results?

- A) Requirement Document
- B) Design Document
- C) Test Summary Report**
- D) User Manual

Answer: C) Test Summary Report

5. What is the main benefit of using testing tools like Bugzilla and TestLink?

- A) Increase Coding Errors
- B) Reduce Software Features
- C) Improve Testing Efficiency and Traceability**
- D) Eliminate Testing

Answer: C) Improve Testing Efficiency and Traceability

Conclusion / Outcome

- Understood the purpose and workflow of **Bugzilla** for defect tracking and **TestLink** for test case management.
- Created and executed test cases using TestLink.
- Learned how to report, assign, and track software defects using Bugzilla.
- Generated **Test Execution Reports** and **Test Summary Reports** to analyze testing progress and results.
- Gained practical knowledge of industry-standard testing tools used in software quality assurance.